

Day -9 : Firewall Basic Theory

1. Block Outbound Traffic for Spotify

1. Open Advanced Security Settings:

Press Win + R, type wf.msc, and press **Enter** to open **Windows Defender Firewall with Advanced Security**.

2. Initiate New Outbound Rule:

Click **Outbound Rules** in the left sidebar, then click **New Rule...** under the **Actions** pane on the right.

3. Select Program Type:

Select **Program** as the Rule Type and click **Next**.

4. Locate Spotify Executable:

Select **This program path**, click **Browse**, and navigate to the Spotify executable (typically located at %APPDATA%\Spotify\Spotify.exe). Click **Next**.

5. Set Action to Block:

Select **Block the connection** and click **Next**.

6. Apply Profiles & Name Rule:

Check all three profiles (**Domain**, **Private**, **Public**), click **Next**, name the rule **Block Spotify Outbound**, and click **Finish**.

2. Allow Inbound Connections for Chrome on Port 443 :

1. Initiate New Inbound Rule:

In the left pane of **Windows Defender Firewall with Advanced Security**, click **Inbound Rules**, then click **New Rule...** in the right pane.

2. Select Program Type:

Select **Program** as the Rule Type and click **Next**.

3. Select Chrome Executable:

Select **This program path**, browse to C:\Program Files\Google\Chrome\Application\chrome.exe (or Program Files (x86) depending on installation), and click **Next**.

4.Set Action to Allow:

Select **Allow the connection** and click **Next**.

5.Set Profiles:

Select the network profiles you wish to apply (**Domain, Private, Public**) and click **Next**.

6.Restrict Rule to Port 443:

Name the rule **Allow Chrome HTTPS Inbound** and click **Finish**. Locate the newly created rule in the list, double-click it, go to the **Protocols and Ports** tab, set **Protocol type** to **TCP**, set **Local port** to **Specific Ports**, enter 443, and click **OK**.

3. Block Traffic from a Specific IP Address (123.45.67.89) :

1.Start New Inbound/Outbound Rule:

In **Windows Defender Firewall**, select either **Inbound Rules** or **Outbound Rules** depending on the direction you wish to block, then click **New Rule....**

2.Select Custom Rule Type:

Select **Custom** as the Rule Type and click **Next**.

3.Apply to All Programs and Protocols:

Leave **All programs** selected and click **Next**. Leave **Protocol type** as **Any** and click **Next**.

4.Specify IP Address in Scope Tab:

Under **Which remote IP addresses does this rule apply to?**, select **These IP addresses**, click **Add....**, enter 123.45.67.89, click **OK**, and click **Next**.

5.Set Action & Save Rule:

Select **Block the connection**, click **Next**, leave all network profiles checked, name the rule **Block IP 123.45.67.89**, and click **Finish**.

4. Hardware Firewall vs. Software Firewall :

Feature / Aspect	Hardware Firewall	Software Firewall
Where it Sits	A physical device placed between your internet router and the whole local network.	Software installed directly on a single computer or laptop (like Windows Defender).
What it Protects	Protects every single device connected to the network at once (computers, phones, IoT devices).	Protects only the specific computer it is installed on.
Performance Impact	Uses dedicated hardware processors; does not slow down individual computers.	Uses the host computer's RAM and CPU to inspect incoming/outgoing traffic.

Primary Use Cases:

- **Hardware Firewall:** Essential for business offices, schools, or data centers to protect hundreds of devices with a centralized network perimeter defense.
- **Software Firewall:** Ideal for personal laptops, home PCs, or remote workers who need protection when connecting to untrusted public Wi-Fi networks away from home.